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SUPERSEDING
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MILITARY SPECIFICATION

FOOD CUTTER AND MIXER MACHINES, VERTICAL, HIGH SPEED, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers commercial, electric-motor driven, vertical, food cutter and mixer machines.

1.2 Classification. Food cutter and mixer machines shall be of the tilting floor model type and of the style and size specified (see 6.2).

Style I - Two speed operation
- Size 1 - 25 quart bowl minimum capacity.
- Size 2 - 40 quart bowl minimum capacity.

Style II - Single speed operation
- Size 1 - 25 quart bowl minimum capacity.
- Size 2 - 40 quart bowl minimum capacity

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper Reinforced and Plain, for Sealing and Securing
PPP-T-60	-	Tape: Packaging, Waterproof

MILITARY

MIL-P-116	-	Preservation, Methods Of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods
MIL-STD-1474	-	Noise Limits for Army Material

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

UNDERWRITERS LABORATORIES, INC.

UL 62 - Flexible Cord and Fixture Wire
UL 73 - Motor Operated Appliances

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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Inventory Management

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 8 - Commercial Powered Food Preparation Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NFS Building, Ann Arbor, MI 48105.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MGI - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, Ny 10017.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard product. The food cutter and mixer machine shall, as a minimum, be in accordance with the requirements of this document and shall be the manufacturer's standard product with any added features needed to comply with the requirements of this document. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this document shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The food cutter and mixer machine shall comply with the applicable requirements of NSF Standard No. 8 and UL 73.

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3.2.1 Compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the food cutter and mixer machine conforms to the requirements of UL and NSF standards, as applicable.

3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished food cutter and mixer machine, or

(2) A certification for the food cutter and mixer machine issued by NSF under their special one-time contract evaluation-certification service shall be submitted, or

(3) A test report, acceptable to the Contracting Officer with the advice of the Army Surgeon General from an independent testing laboratory, indicating that the food cutter and mixer machine has been tested and conforms to NSF.

3.2.1.2 UL. Acceptable evidence of meeting the requirements of UL 73 shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government indicating that the food cutter and mixer machine has been tested and conforms to UL 73.

3.3 Materials (see 6.3) and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial food cutter and mixer machine provided the completed item complies with all provisions of this document.

3.3.1 Stainless steel. Stainless steel shall conform to any type in the 300 or 400 series as specified in ASTM A 167 or ASTM A 176.

3.4 Design and construction. The machines shall be designed for high-speed cutting and mixing of vegetables, meats, dough, and related foodstuffs. Cutting and mixing operations shall be accomplished by direct-motor-driven rotating knives or mixing attachments. Motors for style I machines shall rotate knives at speeds of 1725 and 3450 RPM and motors for the style II machines shall rotate knives at a minimum speed of 1140 RPM. Motor controls shall be mechanically or electrically interlocked with the bowl cover to prevent rotation of the motor when the cover is raised. The machines shall be of the tilting type and shall be supported by a tubular frame base. The bowl cover shall have provisions for supporting a hand-operated baffle to move product being cut from bowl sidewall to cutters.

3.4.1 Bread dough capacity. The machines shall be capable of mixing the following amount of bread dough within the time specified when tested as specified in 4.4.3.

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Style and size of machine	Minimum weight of bread dough	Maximum-time
Style I, size 1	30 pounds	1-1/2 minutes
size 2	50 pounds	1-1/2 minutes
Style II, size 1	24 pounds	3 minutes
size 2	36 pounds	3 minutes

3.4.2 Bowls. Bowls shall be formed of stainless steel specified in 3.3.1. The bowl bottom shall be supported by the motor housing and the drive shaft shall extend up through the bowl bottom and into the center of the bowl. Replaceable shaft seals and bowl seals to prevent leakage shall be provided where the shaft enters the bowl. When tested as specified in 4.4.1, the bowl capacity shall be not less than shown in 1.2 and there shall be no leakage at the seals. The top edge of bowls shall be horizontally flanged on one side to provide a support for cover hinges and to prevent drippings from falling on controls. The bowls shall have one locking lug located opposite the hinges.

3.4.2.1 Bowl insert. The machines shall have a removable durable plastic, aluminum or stainless steel insert sized to fit over the drive shaft and shaped to the inside contour of the bowl. The top edge shall be flanged and the bottom shall be perforated to allow drainage.

3.4.3 Covers. Bowl covers shall be counterbalanced and shall be stainless steel specified in 3.3.1, aluminum alloy or high impact plastic. Hinge straps and locking lug shall be welded to, or integrally cast as a part of the cover. The cover rim shall be formed to fit the bowl rim and shall be fitted with a replaceable gasket. Locking lug shall fit the locking clamp and shall seal the cover to the bowl. The cover shall be fitted with a mixing baffle and a means for monitoring the product during operation. The inspection cover shall have a plastic knob for manual rotation.

3.4.4 Mixing baffle. The mixing baffle shall be aluminum alloy, or stainless steel and shall be manually operated. The baffle shall be removable and shall be shaped to the inside contour of the bowl. The top of the baffle shall be fitted with an offset, removable handle with plastic knob.

3.4.5 Electrical requirements. Unless otherwise specified (see 6.2), motors and other electrical components shall be designed for operation on a nominal 208-volt, 60 hertz, 3-phase power source.

3.4.5.1 Motor. The motor shall be totally enclosed, fan cooled, squirrel cage, induction-run and shall be in accordance with NCMA MG-1 standard. Motor shaft shall be corrosion-resisting steel and direction of shaft rotation shall be indicated on the motor housing. Bearings shall be ball or sleeve type and insulation shall be class A or B. Motor horsepower ratings shall be as follows:

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Machine	Horsepower	Speed (RPM)
Style I, size 1	4.5 minimum	1725
	5.5 minimum	3450
size 2	7.5 minimum	1725
	10.0 minimum	3450
Style II, sizes 1 and 2	5.0 minimum	1140

3.4.5.2 Controls. Motor speeds shall be controlled by a manually operated control. The control for style I machines shall have low, high, and off positions. All controls shall be mechanically or electrically interlocked with the bowl cover so access into the bowl cannot be obtained while the motor is rotating (see 4.4.2).

3.4.5.3 Cord. Cord shall comply with UL 62 and shall be UL type "S", rubber-jacketed, four-conductor, with a 4-prong, grounding type plug. Unless otherwise specified (see 6.2), cord length shall be 6 feet. Cord shall be completely wired to the machine and ready for use.

3.4.5.4 Electromagnetic compatibility. When specified (see 6.2), the food cutter and mixer machine shall be designed and equipped for electromagnetic compatibility in accordance with class C3, group III of MIL-STD-461.

3.4.6 Tilting. The machine shall tilt such that full drainage of the bowl is effected. The bowl, cover, motor, and motor speed control shall tilt as one unit. The unit shall be securely held in a vertical or titled position by a manually operated handle with knob located at the right hand trunnion.

3.4.7 Frame. The machines shall be mounted on a carbon steel tubular frame having four legs. Two trunnions shall be welded to the top of the frame to receive the tilting bearings and shafts. Bearings shall be replaceable. Cross braces left to right and front to back shall be welded to form a rigid assembly. The bottom of each leg shall be fitted with a rubber or plastic foot. Two legs shall also be provided with 3-inch rubber-tired casters which will allow the machine to be moved when tipped. The front cross brace shall have a rubber stop to prevent the motor from contacting the metal frame when returning from a tilted position.

3.4.8 Knobs. Handle knobs shall be moulded plastic or hard rubber and shall be removable for replacement.

3.4.9 Knives. Knives shall be a cutlery grade of stainless steel having a hardness between 52 and 58 when measured on the Rockwell C scale (see 4.4.4). Two knives shall comprise a set and shall be mounted on a shaft of stainless steel, aluminum alloy, or durable plastic construction (see 6.2) and both shaft and knives shall be removable. Bottom of shaft shall fit motor shaft at bottom of bowl.

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valve. If only a specific component was removed, proceed to the

3.4.10 Accessories. Shafts shall be of stainless steel, aluminum alloy, or of a durable plastic construction (see 6.2). The following knives and accessories (see 6.1.1) shall be furnished with each machine:

One set (two) regular edge narrow knives with shaft.

One knead/mix shaft.

One set of special tools and wrenches necessary for removal and replacement of knives, shafts, and other accessories.

3.4.10.1 Optional accessories. The following accessories (see 6.1.1) shall be furnished as specified (see 6.2):

One set (two) regular edge wide knives.

One set (two) wave cut narrow knives.

One set (two) wave cut wide knives.

One grating shaft.

3.5 Operation. The machine shall be operated without load as specified in 4.4.2 to determine controls operate as specified in 3.4.5.2 and that motor speeds conform to 3.4.

3.6 Noise limits. The machine shall not exceed 85 decibels A, weighted at the normal operator position when tested as specified in 4.4.6.

3.7 Identification marking. Identification marking shall be permanently and legibly marked directly on the item or on an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved, or applied by photosensitive means. The plate thickness shall be not less than 0.012 inches. The identification plate shall include in addition to the marking requirements of UL 73, the contract number and the National Stock number.

3.8 Finish. Finish of the food cutter and mixer machines shall be in accordance with applicable requirements of NSF Standard No. 8.

3.9 Workmanship. All components and assemblies of the food cutter and mixer machines shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents, and cracks. Castings, molded parts and stampings, if used, shall be free of sand, fins, pits, blow holes and sprues.

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3.9.1 Metal fabrication. Metal used in the fabrication of the frame shall be free from objectionable kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately.

3.9.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter, welds shall be smooth and free of cracks, burn holes, undercuts or incomplete fusion. All scale and flux shall be removed from the finished weld area.

3.9.3 Fastening devices. Threaded fasteners and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be drawn tight. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of machines of one style and size only. The sample unit shall be one machine with accessories. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with document requirements	X	
Finish	Not as specified	X	
Construction and workmanship general	Component missing	X	
	Any burr, sliver, sharp edge, rough die, tool, gouge, trim or grind marks	X	
	Not constructed as specified	X	
	Not fabricated of material specified	X	
	Component split, cracked, or showing delamination	X	
	Any component fractured, buckled, bent, dented, punctured, or malformed	X	
	Any component misplaced or not in alignment	X	
Fastening devices	Missing, cracked, or stripped	X	
	One bolt, nut, screw, or washer missing	X	
	Two or more bolts, nuts, screws, or washers missing	X	
	Fastener loose		X
	Rivet holes not accurately punched	X	
	Rivet not peened properly		X
Welding	Missing	X	
	Not free from rust, scale, paint, grease, or other foreign matter	X	
	Scale and flux deposits not removed		X
	Weld not smooth and free of cracks, burnholes, undercuts, or incomplete fusion		X
Castings, molded parts and stampings	Miscast or incomplete, not free of sand, fins, pits, sprues, and blowholes	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction (applicable to specific components)			
Bowl	Shaft seal missing or not replaceable	X	
	Top rim not formed as specified	X	
	Top flange not properly located to collect drippings	X	
	Locking devices missing	X	
	Locking devices do not clamp cover tightly to bowl	X	
	Bowl insert not perforated	X	
	Bowl insert top edge not flanged	X	
Cover	Gasket missing or not correctly fitted	X	
	Hinge straps not as specified	X	
	Not counterbalanced	X	
	Inspection cover missing, not as specified, or not removable	X	
Mixing baffle	Not aluminum or stainless steel	X	
	Not shaped to bowl contour	X	
	Handle missing or not shaped as specified	X	
	Handle not removable	X	
Electrical requirements	Control not interlocked with cover	X	
	Cord not 4-prong type	X	
Tilting	Positioning lever does not hold unit in position	X	
	Locking pin missing	X	
Frame	Frame cross bracing not as specified	X	
	Trunnion bearing not replaceable	X	
	Foot or casters missing or not as specified	X	
	Rubber motor stop missing or improperly located	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction (applicable to specific components) (cont'd)			
Knobs	Not replaceable	X	
Knives	Not removable from shaft	X	
Identification marking	Not permanently marked on the item or aluminum, brass, or stainless steel plate		X
	Missing, incomplete, not legible		X

4.2.3 End item testing.

4.2.3.1 Operation testing. Each food cutter and mixer machine shall be tested as specified in 4.4.2. Failure of the test shall be cause for rejection of the machine.

4.2.3.2 Performance testing. One food cutter and mixing machine from the lot shall be tested as specified in 4.4.1, 4.4.3, 4.4.5, and 4.4.6 when applicable. Failure of any test shall be cause for rejection of the lot. One hardness reading shall be taken on each knife furnished with the representative production unit for the test specified in 4.4.4. Failure of the test shall be cause for rejection of all the knives.

4.2.4 Packing inspection. An examination shall be made to determine that the preservation, packing, and marking comply with section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Marking (interior and exterior)	Omitted; incorrect; illegible; of improper size location, sequence, or method of application
Preservation	Preservation missing or not as specified
Materials	Any component missing, damaged, defective, or not as specified
Workmanship	Components not snug fit in container (loose)

4.3 Certificate examination. Certificates of compliance, certified test reports, or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the document requirements shall be examined and validated.

4.4 Methods of inspection. Tests specified in 4.4.2 and 4.4.3 shall be performed with the machine connected to an electrical system having the same characteristics specified in the contract. The manufacturer shall provide motor thermal overload protection for these tests.

4.4.1 Bowl capacity test. The bowl shall be filled with water and then emptied into a graduated container to determine compliance with 3.4.2. Any noncompliance with requirements specified shall constitute failure of the test.

4.4.2 Operation test. Controls shall be placed in each of the operating positions and the speed shall be measured in each of those positions. Speeds shall conform to the requirements of 3.4 The control interlock shall be tested to determine conformance with 3.4.5.2. The test shall consist of placing the control in the operating position(s) to determine that the cover cannot be raised while the shaft is rotating, and when the control is placed in the "off" position the shaft rotation shall stop and the cover may be raised. Any noncompliance with requirements specified shall constitute failure of the test.

4.4.3 Dough mixing test.

4.4.3.1 Dough formulation. The dough mixing test shall be conducted in an ambient temperature of 70°F plus or minus 5°F. Flour shall be hard wheat flour, enriched and bleached, with at least a 14-percent protein content. The following ingredients shall be used for this test with percentages of the ingredients by weight based on a flour content of 100 percent.

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Ingredient	Percentage
Flour	100
Water	60
Yeast (compressed)	2
Salt	2.5
Sugar	5
Shortening	5
Dry skim milk	6
Total percentage	180.5

4.4.3.2 Test procedure. A knead/mix shaft shall be locked in operating position on the drive shaft. The proper amount of ingredients necessary to make the amount of bread dough specified in 3.4.1 shall be dumped into the bowl in accordance with the manufacturers recommendations. The control switch shall be placed in low speed and the machine allowed to run not longer than the maximum time specified in 3.4.1. At the end of this time the machine shall be stopped and the bread dough shall be weighed to determine conformance with 3.4.1. The dough-mix shall be judged by the appearance, quality and if possible by the finished baked product. Any noncompliance with requirements specified shall constitute failure of the test.

4.4.4 Knife hardness test. Knives shall be tested for hardness in accordance with ASTM E 18 (see 3.4.9). Any nonconformance with hardness requirements specified shall constitute failure of the test.

4.4.5 Electromagnetic compatibility test. When electromagnetic compatibility is required, the production unit shall be tested by the contractor in accordance with test method CE03, and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461.

4.4.6 Noise test. The machine shall be tested in accordance with MIL-STD-1474 to determine conformance with 3.6. Any nonconformance with the specified noise limit shall constitute failure of the test.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The mixing baffle handle of the machine shall be removed and the nut replaced on the shaft of the mixing baffle.

5.1.1.2 Preservation. All preservation shall be in accordance with the material and method requirements of MIL-P-116 as hereinafter specified.

5.1.1.2.1 Cleaning. All unpainted, uncoated, exposed ferrous metal surfaces, except corrosion-resisting steel surfaces of the food-cutting and mixing machines, detached parts, accessories, and attachments shall be cleaned process C-1 and thoroughly dried.

5.1.1.2.2 Preservatives. All cleaned uncoated ferrous metal surfaces, except food contacting surfaces, shall be coated with type P-2 or P-6 preservative. All cleaned uncoated ferrous metal surfaces which will be in contact with food ingredients shall be coated with type P-14 preservative. All bearings shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

5.1.1.2.3 Protection of preservative. Surfaces, from which preservatives might be removed by contact with packaging materials, shall be covered, bagged, or wrapped with greaseproof barrier material conforming to type I or II, grade A of MIL-B-121. Barrier material shall be secured with twine made of jute, sisal, manila, or cotton, or tape conforming to type III, grade A of PPP-T-45, or heat sealed when heat sealable barrier material is used.

5.1.1.2.4 Sealing. Exterior switches, terminals, control boxes, and all openings in motors shall be sealed with tape conforming to type III or IV, class 1 of PPP-T-60. A caution tag containing the words, "Remove Tape from Electric Motor Prior to Operation of the Machine", shall be secured in a prominent location on the machine.

5.1.1.3 Component packing. Accessories, detached parts, attachments and instruction manual, prepared as specified, shall be placed in a snug-fitting fiberboard or nailed wood box or boxes conforming to style RSC, type CF (variety SW) or type SF, class domestic of PPP-B-636 or class 1, style 2 or 4 of PPP-B-621, respectively. The electric cord shall be secured to the machine with cord, twine, tape or other suitable means. All components that are free to move, including the bowl, shall be secured by appropriate means. The contents shall be packed, cushioned, blocked, or braced within the box in accordance with the manufacturer's standard practice to prevent movement while in transit. Each box shall be securely closed in accordance with the appendix of the box specification.

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* 5.1.2 Commercial. Food cutting and mixing machines shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each complete food cutting and mixing machine, preserved as specified in 5.1, shall be packed in a cleated plywood shipping container conforming to overseas type, type 3 load of PPP-B-601. Lumber blocking and bracing shall be used to provide a 1-inch clearance between the interior surface of the bottom of the shipping container and the bottom of the legs of the machine. Blocking and bracing, with lumber, anchoring, and waterproofing shall be in accordance with MIL-STD-1186. Each shipping container, shall be provided with skids in accordance with PPP-B-601.

5.2.2 Level B packing. Each complete food cutting and mixing machine, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except that the shipping container shall be domestic type and waterproofing shall not be required.

5.2.3 Commercial packing. Food cutting and mixing machines, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The machines are intended for use in kitchens and hospitals for high speed food processing.

6.1.1 Accessory uses. Accessories specified in 3.4.10 and 3.4.10.1 are primarily intended for the following uses:

Narrow knives	-	Chopping meat and vegetables
Wide knives	-	Cutting frozen foods
Wave cut knives	-	Chopping oily food such as nuts
Grating shaft	-	Grating hard vegetables and mixing
Kneading and mixing shaft	-	Kneading yeast doughs, blending, mixing, cake batters, or mayonnaise

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Style and size of machine required (see 1.2).
(c) Voltage, frequency and phase if other than specified (see 3.4.5).
(d) Electrical cord length if different (see 3.4.5.3).
(e) When electromagnetic compatibility is required (see 3.4.5.4).
(f) Shaft Material (see 3.4.9 and 3.4.10).
(g) Additional accessories required (see 3.4.10.1)
(h) Selection of the applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.4 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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